Benchmarking OC25 and UMA foundational models for organic molecules on Pd(111) energetics against DFT and correcting using delta learning

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Abstract: This project benchmarks pretrained OC25 and UMA models against RPBE and PBE DFT energies for organic molecules and organic adsorbates on Pd(111). The comparison includes molecular, slab–molecule, and adsorption energies across a chemically diverse set of systems. OC25 follows the DFT adsorption-energy trends more closely than UMA, which strongly underpredicts adsorption strength. A simple linear delta-learning correction further improves OC25, reducing the RPBE adsorption-energy MAE from 0.794 eV to 0.060 eV. These results suggest that OC25 is the stronger baseline model for organic–Pd(111) adsorption energetics in this dataset.

Keywords: MLIP, DFT, adsorption, machine learning, interatomic potentials

Introduction

Machine learning interatomic potentials (MLIPs) are becoming an important tool in computational catalysis because they can approach density functional theory (DFT) accuracy at a much lower computational cost[1]. This is especially valuable for catalytic interfaces involving metal surfaces and organic adsorbates, where direct DFT screening of many adsorbate combinations is expensive. In this project, I focus on benchmarking pretrained MLIP energy predictions for organic molecules and organic molecules adsorbed on a Pd(111) surface.

The project scope focuses on a controlled comparison between OC25 and UMA model[2] predictions against DFT values computed with RPBE and PBE functionals. I will also show results for a finetuned DPA model[3,4] with OC25 dataset. This scope made it possible to compare absolute molecule energies, slab–molecule energies, and adsorption energies for a larger set of organic molecules. The main scientific question is whether pretrained foundation-style models can reproduce the DFT energetic trends for organic adsorbates on a metal surface and whether simple calibration can reduce systematic errors.

The systems (dataset) include individual organic molecules such as benzaldehyde, phenol, catechol, guaiacol, furoic acid, furfural, toluene, and xylene, along with paired systems where benzaldehyde is combined with a second adsorbate. These molecules are relevant to biomass-derived catalytic chemistry and provide a chemically diverse benchmark containing aromatic, oxygenated, and substituted species. By comparing MLIP values directly to RPBE and PBE references, this project evaluates both model transferability and the size of model-dependent energetic bias.

Prior Work

Recent years have seen rapid progress in MLIPs for chemistry and materials science. Large pretrained models trained on broad DFT datasets have shown promising performance across molecules, bulk materials, and surfaces. Equivariant graph neural networks, Atomic Cluster Expansion-based models[5], and transformer-style atomistic models now provide practical alternatives to repeatedly training system-specific force fields. However, many benchmarks still emphasize gas-phase molecules, bulk crystals, or clean surfaces, while organic adsorption on transition-metal surfaces remains a difficult transferability test.

This project builds on that literature by comparing two pretrained model families. UMA-S-1p2 is a universal model for atoms trained across broad materials and molecular data[2,6]. OC25/eSEN is used here as a FairChem pretrained model intended for catalytic and surface chemistry settings. Both are evaluated with the same post-processing workflow so that differences in performance can be interpreted as model behavior rather than workflow artifacts.

The DFT values provide two reference levels. RPBE is used as the main adsorption-energy reference because it is commonly used for surface adsorption studies, while PBE provides a second comparison for molecular and slab-containing energy trends. The use of two DFT functionals is important because agreement with one reference does not guarantee agreement with another; instead, the benchmark measures how consistently each MLIP follows the relative ordering and magnitude of DFT energies across related organic systems.

Method

The benchmark was organized around three energy comparisons. First, absolute molecular energies were compared across DFT PBE, DFT RPBE, OC25, and UMA. Second, slab-plus-molecule energies were compared for Pd(111)-containing systems. Third, adsorption energies were calculated and compared against RPBE and PBE references. The adsorption-energy comparison is the most chemically relevant quantity because it removes much of the absolute-energy offset and focuses on the interaction between the organic molecule and the surface.

The workflow used Python-based data analysis and visualization tools, with structures and energy bookkeeping organized consistently across models. For each system, predicted model energies were matched to the corresponding DFT reference. Parity plots were then generated to show whether model predictions follow the ideal one-to-one relationship with DFT. Energy-versus-system plots were used to inspect whether the models reproduce trends across chemically related molecules and benzaldehyde-containing coadsorption systems.

In addition to raw model predictions, a simple delta-learning correction was tested for adsorption energies. The correction used RPBE adsorption energies as the target and either OC25 or UMA predictions as the baseline. Three cases were compared: the raw baseline, a constant-bias correction, and a linear calibration. This calibration step does not represent full MLIP fine-tuning; instead, it tests whether a small data-driven correction can remove systematic model bias in the adsorption-energy predictions.

Results

For molecular energies, both OC25 and UMA follow the qualitative DFT trends across the organic molecule set (**Figure 4**), but the agreement differs by model and by energy type. The adsorption energy plot (**Figure 4**) show that the models reproduce the ordering of many low- and high-energy systems, indicating that the pretrained representations capture meaningful chemical variation.

For adsorption energies against RPBE, OC25 performs substantially better than UMA in the raw baseline comparison (**Figure 1**). Across 35 adsorption systems, the mean absolute OC25 error relative to RPBE is 0.8 eV, while the mean absolute UMA error is 3.3 eV (**Figure 2**). OC25 generally predicts adsorption energies that are too weak compared with RPBE, but the error is closer to a consistent offset. UMA predictions are much closer to zero adsorption energy over many systems, which causes large underbinding relative to RPBE.

The benzaldehyde coadsorption systems provide a clear example of this behavior. RPBE adsorption energies for these paired systems are typically between about -4 and -5.4 eV. OC25 predicts the same systems at roughly -3 to -4.2 eV, preserving the broad trend but underestimating adsorption strength. UMA often predicts values near zero to about -0.6 eV, which loses much of the adsorption magnitude. This difference suggests that OC25 transfers more directly to the organic–Pd(111) adsorption-energy task than UMA in the zero-shot setting.

The simple delta-learning test further supports this interpretation (**Figure 3**). For OC25, a linear calibration reduces the adsorption-energy MAE from 0.8 eV to 0.06 eV and the RMSE from 0.8 eV to 0.07 eV. A constant-bias correction also improves OC25, reducing the MAE to 0.272 eV. For UMA, the raw MAE is 3.271 eV, and linear calibration reduces it to 0.572 eV. This is a major improvement, but the calibrated UMA error remains much larger than calibrated OC25. Overall, OC25 appears to provide the stronger baseline for this dataset, especially when paired with a small linear correction.

Conclusions

This project compared pretrained OC25 and UMA model predictions against RPBE and PBE DFT reference values for organic molecules and organic adsorbates on Pd(111). The results show that both models capture some broad energetic trends for the molecule set, but their accuracy differs strongly for adsorption energies. OC25 provides the better zero-shot baseline, with adsorption energies that are systematically weaker than RPBE but still close enough to preserve the main trends across many systems. UMA shows much larger underbinding for the adsorption systems, often predicting adsorption energies close to zero when RPBE gives strongly negative values.

A simple delta-learning calibration substantially improved the OC25 adsorption-energy predictions, reducing the MAE from 0.794 eV to 0.060 eV. UMA also improved after calibration, but its corrected error remained larger than OC25. Overall, these results

suggest that OC25 transfers more reliably to organic–Pd(111) adsorption energetics in this dataset, especially when combined with a small data-driven correction. Future work could extend this benchmark to larger structural datasets, force comparisons, and molecular dynamics tests once more DFT reference data are available.

Supplemental Resources

The final documents for the project are uploaded to GitHub Repository[7].

References

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Figures

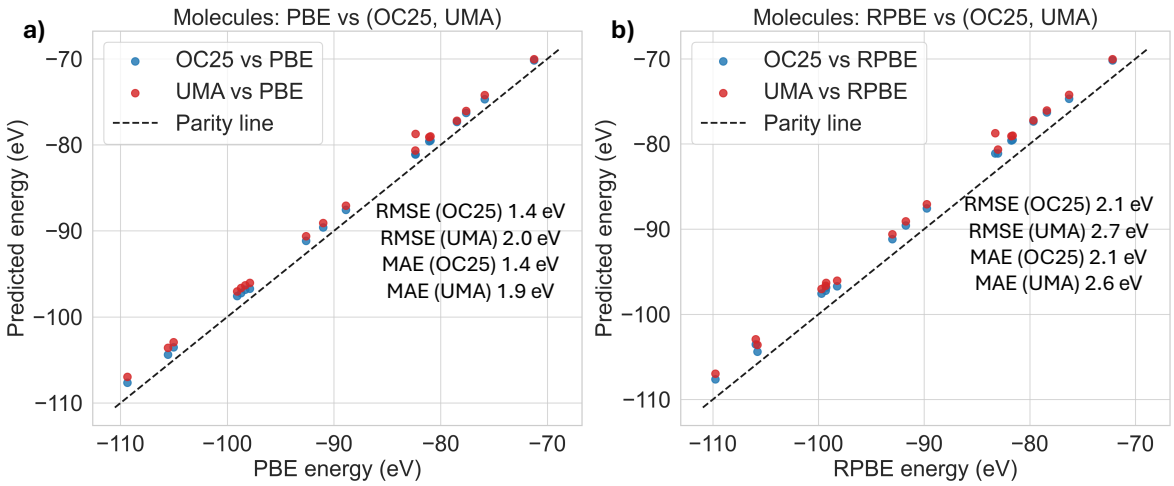

Figure 1: Parity comparison of molecule energies from **(a)** PBE and **(b)** RPBE with OC25 and UMA predictions. OC25 is closer to the one-to-one line, while UMA strongly underestimates adsorption strength for many systems.

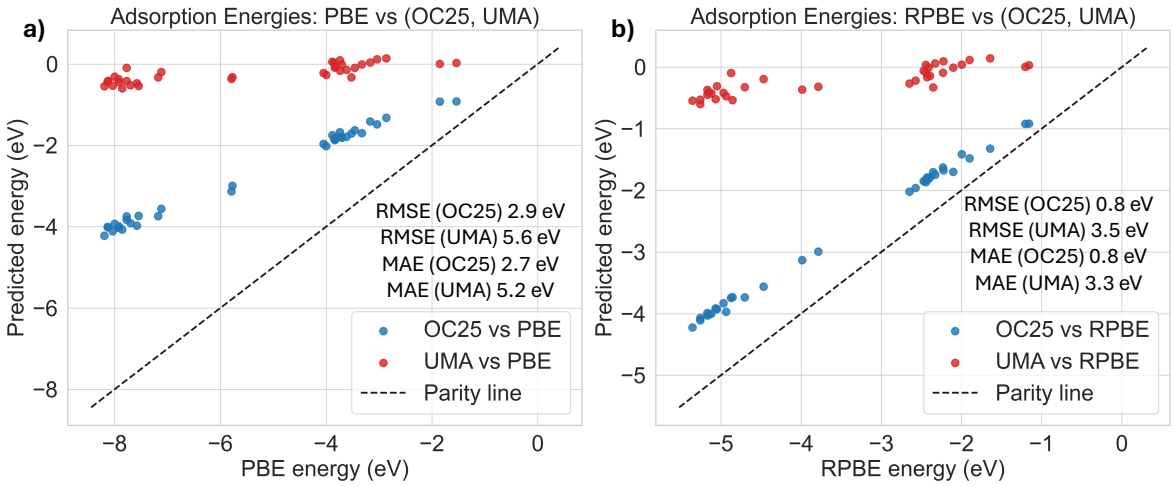

Figure 2: Parity comparison of adsorption energies from **(a)** PBE and **(b)** RPBE with OC25 and UMA predictions. OC25 is closer to the one-to-one line, while UMA strongly underestimates adsorption strength for many systems.

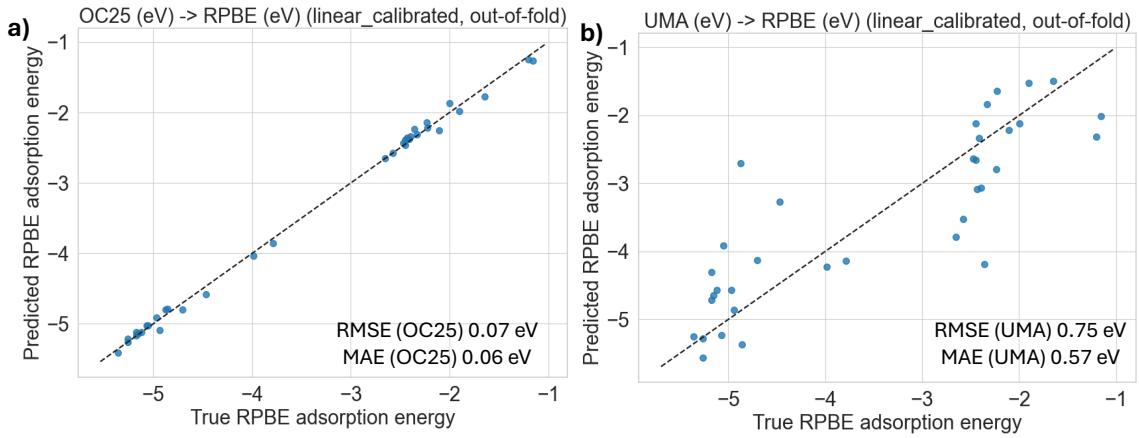

Figure 3: Delta-learning calibration for adsorption energies relative to RPBE for **(a)** OC25 and **(b)** UMA. Linear calibration greatly reduces systematic error, especially for OC25, whose MAE decreases from 0.794 eV to 0.060 eV.

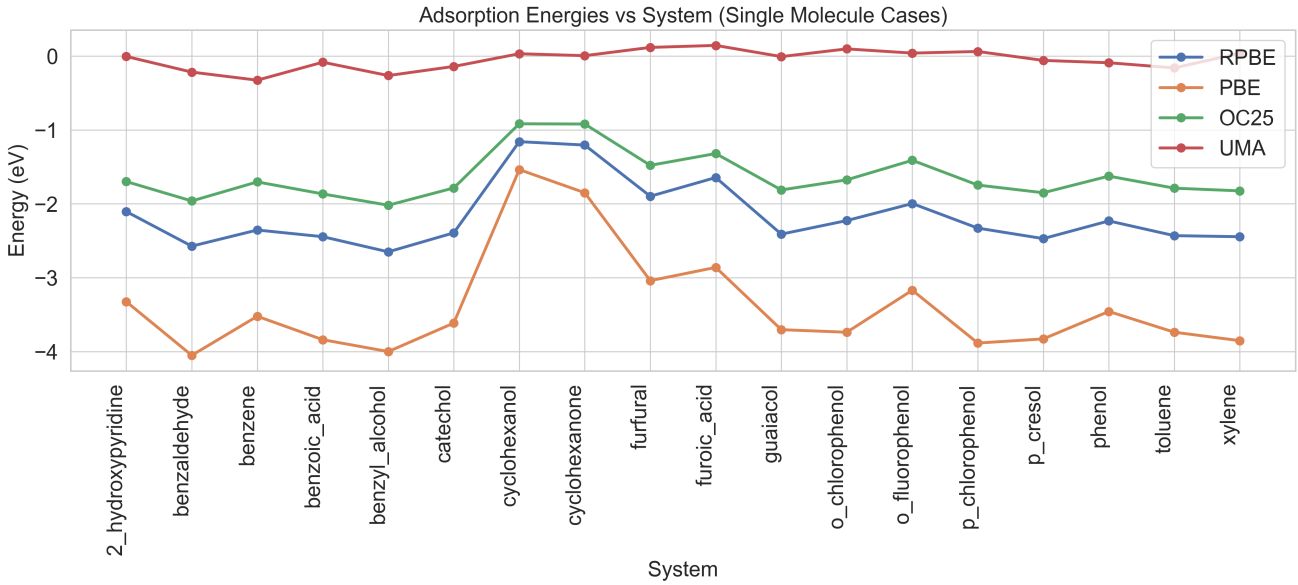

Figure 4: Adsorption energies for the organic molecule on Pd(111) using DFT PBE and RPBE and OC25 and UMA.